

Positive real asymptotic inversion

Power–logarithmic series, exact irrational supports,
and a Wolfram Language package

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Abstract

The real inverse of $x + x^2(1 + \log x)$ at 0^+ is not a Taylor series, and the inverse of $x + x^{\sqrt{2}}$ does not have a rational Puiseux support. Both nevertheless admit a systematic, branch-explicit inversion calculus. We develop it for every exact finite model

$$f(x) = ax^p + \sum_{j=1}^m x^{p+\eta_j} P_j(\log x), \quad a, p, \eta_j > 0, \quad P_j \in \mathbb{R}[L].$$

After the positive substitution $v = x^p$, a local Lagrange–Bürmann formula gives every inverse coefficient as a finite product of first-order operators on logarithmic polynomials. The output powers lie on a finitely generated positive real semigroup, so irrational exponents and resonances can be handled exactly. We prove branch existence and uniqueness, convergence of the expansion for sufficiently small positive arguments, and a next-weight remainder theorem that retains necessary powers of $\log y$. An independent composition procedure checks the computed coefficients, with the correct residual-to-inverse error conversion for $p \neq 1$. The accompanying package implements this finite-model theory, exposes explicit remainder scales, and supplies a more general derivative-form helper without pretending to certify its analytic hypotheses.

Deliverables	This article and its editable source; commented package; worked examples; native Wolfram tests; independent verification scripts.
Verification	66 independent mathematical and static checks passed, including numerical comparisons at 180 decimal digits. The supplied native Wolfram tests were not executed: the available service returned HTTP 404 and no local Wolfram kernel was installed.

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1 The question and its concrete answers

The archived question asks for the positive real inverse near zero of

$$f(x) = x + x^2(1 + \log x), \quad x > 0, \quad (1.1)$$

and subsequently asks for the analogous expansion of

$$f_\alpha(x) = x + x^\alpha, \quad \alpha = \sqrt{2}. \quad (1.2)$$

The archive contains the question in Markdown and PDF; the Markdown preserves both examples and their mathematical notation. The public question is the corresponding primary source [1]. Its displayed computer outputs are historical reports by the questioner, not results reproduced in a current Wolfram kernel for this article.

Put $L = \log y$. The first answer, through four correction orders, is

$$\begin{aligned} f^{-1}(y) = & y - y^2(1 + L) + y^3(2L^2 + 5L + 3) \\ & - y^4 \left(5L^3 + \frac{41}{2}L^2 + 27L + \frac{23}{2} \right) \\ & + y^5 \left(14L^4 + \frac{241}{3}L^3 + \frac{335}{2}L^2 + 151L + \frac{299}{6} \right) + O(y^6(1 + |L|)^5). \end{aligned} \quad (1.3)$$

In particular, the correct two-block statement is

$$f^{-1}(y) = y - y^2(1 + \log y) + O(y^3(1 + |\log y|)^2). \quad (1.4)$$

The error is *not* $O(y^3)$: its next coefficient is a nonconstant quadratic polynomial in $\log y$. More sharply,

$$\frac{f^{-1}(y) - y + y^2(1 + \log y)}{y^3(\log y)^2} \longrightarrow 2.$$

This distinction matters even when a computer algebra system displays an informal series-order marker that appears to suppress logarithms.

The second answer is exactly the expansion anticipated in the question:

$$f_{\sqrt{2}}^{-1}(y) = y - y^{\sqrt{2}} + \sqrt{2} y^{2\sqrt{2}-1} - \frac{6 - \sqrt{2}}{2} y^{3\sqrt{2}-2} + O(y^{4\sqrt{2}-3}). \quad (1.5)$$

The next, explicitly computable term is

$$\frac{17\sqrt{2} - 12}{3} y^{4\sqrt{2}-3}.$$

All powers in these statements are evaluated on the positive real axis. There is no need to choose an unspecified complex inverse and subsequently try to repair its branch by assumptions on the output variable.

1.1 Immediate package usage

After extracting the archive, use a path to the package file:

```
Get["Kernel/RealInverseAsymptotics.wl"];
Clear[x, y, z];

r = RealInverseExpansion[
  x + x^2 (1 + Log[x]), {x, y}, 6];
r["Expression"]
r["RemainderScale"]
InverseResidual[r]["VanishingBelowCutoff"]

s = RealInverseExpansion[
  x + x^Sqrt[2], {x, z}, 4 Sqrt[2] - 3];
s["Expression"]
s["RemainderScale"]
```

The cutoff is *exclusive*: the first call keeps output powers strictly below 6, and the second keeps exactly the four blocks displayed in (1.5). The remainder field is a scale for a mathematical big- O statement, not a measured numerical error and not a substituted `SeriesData` object. The residual call is designed to return `True`; native-kernel execution status is recorded separately in Section 11.

1.2 What is classical, and what is assembled here

The coefficient identity used below is classical Lagrange inversion; see the NIST DLMF [2] and Gessel's survey, especially Theorem 2.1.1 [3]. We do not claim to have invented that identity. The contribution here is its complete adaptation to the specific nonanalytic endpoint problem: a real branch condition, a positive-core normalization, an exact irrational support calculus, a finite coefficient algorithm, a proved logarithm-sensitive remainder, and an implementation with an independent residual calculation.

The supplied repository's group README explicitly distinguishes the near-identity operator form of Lagrange–Bürmann from the classical coefficient form and cautions against identifying results merely because they share terminology [4]. The derivation in Section 4 makes the bridge explicit. The repository's canonical-volume README also separates exact, partial, and absent formalization claims [5]. No Lean verification of the present package or of this article is asserted. Repository context consulted here consists of those two READMEs, not a line-by-line review of the complete consolidated volumes.

2 The model, scales, and finite supports

2.1 Exact input class

Our main class is

$$f(x) = ax^p + \sum_{j=1}^m x^{p+\eta_j} P_j(\log x), \quad a > 0, \quad p > 0, \quad \eta_j > 0, \quad P_j \in \mathbb{R}[L]. \quad (2.1)$$

We first combine equal input powers and remove zero polynomials. The sum is finite; the η_j need not be integers, rational numbers, or rationally independent. Coefficients of the P_j can have either sign. The leading coefficient a is constant, rather than a polynomial in $\log x$. A pure monomial, with $m = 0$, is included and has the exact inverse $(y/a)^{1/p}$. Statements below involving a minimum correction weight or a first excluded weight assume $m \geq 1$; they are unnecessary in the pure-monomial case.

The model is initially regarded as an *exact function*. An input asymptotic expansion with an unknown remainder is a different object; Section 8.4 explains how that remainder affects the inverse. Omitting it silently would turn a correct calculation for the model into an unjustified assertion about the original function.

Define the normalization

$$t = \frac{y}{a}, \quad v = x^p, \quad q = \frac{1}{p}, \quad \delta_j = \frac{\eta_j}{p}, \quad Q_j(L) = \frac{1}{a} P_j\left(\frac{L}{p}\right). \quad (2.2)$$

Then the inverse problem is

$$t = v + \psi(v), \quad \psi(v) = \sum_{j=1}^m v^{1+\delta_j} Q_j(\log v), \quad x = v^q. \quad (2.3)$$

The positive-real identities $\log(x^p) = p \log x$ and $x = (x^p)^{1/p}$ justify this substitution. They are branch assertions, not unrestricted rules for complex powers.

2.2 Power–logarithmic blocks

A block is an expression $t^\gamma P(\log t)$, with P a polynomial. It is convenient to use the positive logarithmic envelope

$$\ell(t) = 1 + |\log t|, \quad 0 < t < 1. \quad (2.4)$$

For every $\varepsilon > 0$ and every fixed nonnegative integer d ,

$$t^\varepsilon \ell(t)^d \longrightarrow 0. \quad (2.5)$$

Indeed, putting $s = -\log t$ gives $e^{-\varepsilon s}(1+s)^d \rightarrow 0$. Consequently, an increase in the power of t dominates any fixed change in logarithmic degree. Within a single power, the highest nonzero power of $\log t$ dominates the lower ones in magnitude.

A truncation by powers should retain entire coefficient polynomials. For example, $y^3(2L^2 + 5L + 3)$ is one block, but three scalar monomials. A requested number of “terms” is therefore ambiguous unless the grading is specified. The package uses an exclusive real-power cutoff rather than an ambiguous term count.

Lemma 2.1 (Uniqueness of a leading block). *Suppose $\gamma_1 < \dots < \gamma_r$ and $P_1 \neq 0$. The finite sum $\sum_i t^{\gamma_i} P_i(\log t)$ is asymptotic to $t^{\gamma_1} P_1(\log t)$ as $t \rightarrow 0^+$. In particular, distinct powers cannot cancel each other’s leading asymptotic behavior.*

Proof. A nonzero polynomial satisfies $P_1(L) \sim cL^d$ as $L \rightarrow -\infty$. Divide the sum by the first block and apply (2.5) to every remaining ratio. $\square$

2.3 Why irrational exponents cause no finite-truncation problem

Let

$$\Gamma = \{\mathbf{k} \cdot \boldsymbol{\delta} : \mathbf{k} = (k_1, \dots, k_m) \in \mathbb{N}_0^m\}. \quad (2.6)$$

The inverse support will be contained in $q + \Gamma$. The group generated by irrationally related real numbers can be dense; that observation is irrelevant to the *positive semigroup* used here.

Lemma 2.2 (Local finiteness and finite fibers). *For every finite B , there are only finitely many $\mathbf{k} \in \mathbb{N}_0^m$ with $\mathbf{k} \cdot \boldsymbol{\delta} \leq B$. In particular, $\Gamma \cap [0, B]$ is finite, and each exponent has only finitely many multi-index representations.*

Proof. Writing $\delta_* = \min_j \delta_j > 0$ gives $|k|\delta_* \leq k \cdot \delta \leq B$. Alternatively, each coordinate satisfies $k_j \leq B/\delta_j$. Either inequality puts the relevant integer vectors in a finite box. $\square$

Different vectors can still give the same power. Such a collision is called a resonance here. It must be resolved by *adding* the corresponding logarithmic polynomials. Treating vectors as distinct output powers would be mathematically wrong. Exact comparisons are essential near resonances: rounding an exponent is not an acceptable way to decide whether a term belongs below a cutoff.

3 Selecting the positive real branch

Theorem 3.1 (Existence, uniqueness, and leading behavior). *For every model (2.1), there is $\varepsilon > 0$ such that f is strictly increasing and positive on $(0, \varepsilon)$ and maps that interval onto $(0, f(\varepsilon))$. Its inverse $x(y)$ on this interval is the unique positive branch satisfying*

$$\frac{x(y)}{(y/a)^{1/p}} \longrightarrow 1 \quad (y \rightarrow 0^+). \quad (3.1)$$

In the normalized coordinate, $v(t)/t \rightarrow 1$.

Proof. Write $f(x) = ax^p(1 + r(x))$, where

$$r(x) = \frac{1}{a} \sum_j x^{\eta_j} P_j(\log x).$$

By (2.5), $r(x) = o(1)$ and

$$xr'(x) = \frac{1}{a} \sum_j x^{\eta_j} (\eta_j P_j + P'_j)(\log x) = o(1).$$

Therefore

$$f'(x) = apx^{p-1} \left(1 + r(x) + \frac{xr'(x)}{p} \right) = apx^{p-1} (1 + o(1)) > 0 \quad (3.2)$$

for sufficiently small positive x . Also $f(x) > 0$ there and $f(x) \rightarrow 0$. Continuity and strict monotonicity give the inverse interval. Substitute $x = x(y)$ in $f(x)/(ax^p) \rightarrow 1$ to obtain $x(y)^p/(y/a) \rightarrow 1$. The positive p th root gives (3.1). Any other positive branch tending to zero eventually lies in the same strictly monotone interval and is identical to it. $\square$

The theorem does not assert global invertibility for every model. Coefficients of the corrections may create critical points farther from zero. The asymptotic condition chooses a local real germ. A global inverse exists for both examples in the question, as will be checked explicitly.

3.1 Why assumptions on a complex inverse can miss this branch

The ordinary analytic inverse theorem is centered at a point where the forward map is analytic and has nonzero derivative; the DLMF states those hypotheses explicitly [2]. In (1.1), $x = 0$ is not an analytic point of $\log x$. A Taylor expansion about a nonzero complex solution of $f(x) = 0$ solves a *different* local inverse problem.

For example, on compatible logarithmic sheets the nonzero equation $x(1 + \log x) = -1$ leads to

$$x = \exp(-1 + W_k(-e)).$$

An expansion with such a nonzero constant term cannot satisfy $x(y) \sim y$. The correct branch condition is therefore built into the coordinate normalization, rather than entrusted to an unspecified `InverseFunction` representation. The current documentation describes `InverseFunction` and restrictions of its domain [6]; this article makes no claim that the historical calls in the question still behave identically in every current release.

4 Lagrange inversion away from the singular endpoint

4.1 A local auxiliary parameter

Fix $t > 0$, temporarily. The function ψ in (2.3) is analytic on a sufficiently small complex disk around this *positive* point, after choosing the logarithm continued from its real value. Introduce an auxiliary parameter s :

$$V = t - s\psi(V), \quad V(t, 0) = t. \quad (4.1)$$

This equation is regular at $s = 0$, even though ψ is not analytic at $t = 0$. Analyticity in s is the useful regularity.

Lemma 4.1 (Derivative-form Lagrange–Bürmann). *For an observable Φ analytic near the fixed positive t ,*

$$\Phi(V(t, s)) = \Phi(t) + \sum_{n \geq 1} \frac{(-s)^n}{n!} \frac{d^{n-1}}{dt^{n-1}} (\Phi'(t)\psi(t)^n) \quad (4.2)$$

for sufficiently small $|s|$.

Proof. Set $w = V - t$. Then $w = sR(w)$ with $R(w) = -\psi(t + w)$. The classical coefficient form of Lagrange inversion gives

$$[s^n]\Phi(t + w) = \frac{1}{n} [z^{n-1}]\Phi'(t + z)R(z)^n, \quad n \geq 1.$$

This is precisely the form in Gessel’s Theorem 2.1.1 [3]. Taylor coefficient extraction at $z = 0$ turns the right side into

$$\frac{(-1)^n}{n!} \frac{d^{n-1}}{dz^{n-1}} (\Phi'(t + z)\psi(t + z)^n) \Big|_{z=0}.$$

Differentiation in the translated variable gives (4.2). The analytic inverse theorem for $w - sR(w) = 0$ justifies the convergent local identity in s . Notice that t was a positive parameter throughout; no Taylor theorem was applied at the logarithmic singularity. $\square$

For $\Phi(v) = v$, the formula becomes

$$V(t, s) = t + \sum_{n \geq 1} \frac{(-s)^n}{n!} \left(\frac{d}{dt} \right)^{n-1} \psi(t)^n. \quad (4.3)$$

For the original inverse we need $\Phi(v) = v^q$, which gives

$$V(t, s)^q = t^q + \sum_{n \geq 1} \frac{q(-s)^n}{n!} \left(\frac{d}{dt} \right)^{n-1} (t^{q-1}\psi(t)^n). \quad (4.4)$$

The factor q and the shift $q - 1$ are important. Computing an inverse for v and then treating it as the inverse for x would lose both.

The local lemma alone does not yet permit setting $s = 1$ while $t \rightarrow 0$. A uniform smallness argument is still needed. Section 6 provides it, rather than silently identifying a formal perturbation series with an asymptotic expansion of the actual inverse.

4.2 Differentiating a logarithmic block

Let $\mathcal{D} = d/dL$ act on coefficient polynomials.

Lemma 4.2 (Block differentiation). *For any real λ , polynomial P , and integer $r \geq 0$,*

$$\left(\frac{d}{dt}\right)^r (t^\lambda P(\log t)) = t^{\lambda-r} \prod_{j=0}^{r-1} (\mathcal{D} + \lambda - j) P(L) \Big|_{L=\log t}. \quad (4.5)$$

An empty product is the identity operator.

Proof. The first derivative is $t^{\lambda-1}(\lambda P + P')(\log t)$. Apply this identity successively. The exponent decreases by one at each step, while the constant-coefficient polynomial differential operators commute. $\square$

Differentiation with respect to t therefore requires only polynomial arithmetic in L . This is the central computational simplification: logarithms are neither discarded nor treated as constants when the independent variable is differentiated.

5 The coefficient theorem

For $\mathbf{k} \in \mathbb{N}_0^m \setminus \{0\}$ write

$$n = |\mathbf{k}| = \sum_j k_j, \quad \beta = \mathbf{k} \cdot \delta, \quad \mathbf{k}! = \prod_j k_j!, \quad Q^{\mathbf{k}}(L) = \prod_j Q_j(L)^{k_j}. \quad (5.1)$$

Define

$$B_{\mathbf{k}}^{(q)}(L) = \frac{q(-1)^n}{\mathbf{k}!} \prod_{r=1}^{n-1} (\mathcal{D} + \beta + q + r) Q^{\mathbf{k}}(L). \quad (5.2)$$

For $n = 1$ this says $B_{\mathbf{e}_j}^{(q)} = -qQ_j$.

Theorem 5.1 (Complete finite-model coefficient formula). *The positive inverse of (2.1) has the expansion*

$$x(y) = t^q + \sum_{\mathbf{k} \neq 0} t^{q+\mathbf{k} \cdot \delta} B_{\mathbf{k}}^{(q)}(\log t), \quad t = y/a. \quad (5.3)$$

Each coefficient is a polynomial. Equal powers are combined by

$$C_{\beta}(L) = \sum_{\mathbf{k} \cdot \delta = \beta} B_{\mathbf{k}}^{(q)}(L). \quad (5.4)$$

Both the aggregate at a fixed power and the set of terms below a fixed power cutoff are finite. The series converges as a sum of blocks for sufficiently small $t > 0$, and has the asymptotic remainder described in Theorem 6.2.

Coefficient calculation. The multinomial theorem gives

$$\psi(t)^n = \sum_{|\mathbf{k}|=n} \frac{n!}{\mathbf{k}!} t^{n+\beta} Q^{\mathbf{k}}(\log t).$$

Insert this expression in (4.4). The initial exponent in the differentiated block is $q - 1 + n + \beta$. Lemma 4.2, with $r = n - 1$, leaves the power $q + \beta$ and produces the constants $q + \beta + 1, \dots, q + \beta + n - 1$. The multinomial $n!$ cancels the Lagrange denominator $n!$. This proves (5.2); Lemma 2.2 proves all stated finiteness claims. The analytic and remainder assertions are proved in the next section. $\square$

5.1 Degree and leading-logarithm checks

Let $d_j = \deg Q_j$, and let c_j be its leading coefficient. For a nonzero multi-index block,

$$\deg B_{\mathbf{k}}^{(q)} = \sum_j k_j d_j, \quad (5.5)$$

$$[L^{\sum_j k_j d_j} B_{\mathbf{k}}^{(q)}] = \frac{q(-1)^n}{\mathbf{k}!} \prod_{r=1}^{n-1} (\beta + q + r) \prod_j c_j^{k_j}. \quad (5.6)$$

Every constant $\beta + q + r$ is positive, so the highest-degree term cannot disappear inside one nonzero multi-index block. It can disappear when several resonant blocks are added. This supplies an exact internal check and explains why a conservative remainder degree should be computed before assuming any cancellation.

One may expand the operator product using elementary symmetric polynomials:

$$\prod_{r=1}^{n-1} (\mathcal{D} + c_r) P = \sum_{j=0}^{\min(n-1, \deg P)} e_{n-1-j}(c_1, \dots, c_{n-1}) P^{(j)}. \quad (5.7)$$

The implementation uses repeated first-order applications instead. That approach avoids constructing an extra family of symmetric-polynomial expressions and works directly in the coefficient polynomial ring.

5.2 An alternative recursive construction

The normalized equation can also be written with $v = t(1 + u)$:

$$0 = u + \sum_j t^{\delta_j} (1 + u)^{1+\delta_j} Q_j(L + \log(1 + u)), \quad L = \log t. \quad (5.8)$$

The derivative of its left side with respect to u has constant term 1. Consequently the coefficient at each new support weight is determined linearly by previously determined coefficients. Reversion by coefficient comparison is therefore triangular, not a large nonlinear solve.

Formula (5.2) solves this triangular process in a finite nonrecursive form. The verification script uses the recursive unit-series identities as an independent check; the package's residual routine uses a third viewpoint, direct composition of a finite candidate.

6 Convergence and logarithm-sensitive remainders

6.1 A uniform contraction in the auxiliary parameters

Assume $m \geq 1$, and put

$$\delta_* = \min_j \delta_j, \quad d_* = \max_j d_j, \quad \epsilon(t) = t^{\delta_*} \ell(t)^{d_*}. \quad (6.1)$$

Thus $\epsilon(t) \rightarrow 0$. Constants below may depend on the fixed model, but not on sufficiently small positive t .

Proposition 6.1 (Convergent perturbation expansion). *The series (5.3) converges absolutely as a sum of its multi-index block values for sufficiently small $t > 0$. Its truncation by total perturbation degree satisfies, for each fixed $N \geq 0$,*

$$x(y) - t^q - \sum_{1 \leq |\mathbf{k}| \leq N} t^{q+\mathbf{k} \cdot \delta} B_{\mathbf{k}}^{(q)}(\log t) = O\left(t^q \epsilon(t)^{N+1}\right). \quad (6.2)$$

Proof. Introduce a separate complex parameter s_j for each correction in (5.8). On the closed disk $|u| \leq 1/2$, the functions $(1+u)^{1+\delta_j}$ and $\log(1+u)$ are analytic and bounded, as are the derivatives needed below. Polynomial growth gives uniform bounds

$$|t^{\delta_j}(1+u)^{1+\delta_j}Q_j(L+\log(1+u))| \leq C\epsilon(t)$$

and the same type of bound for the derivative in u . Choose a fixed K large enough that, whenever

$$|s_j| \leq R(t) := \frac{1}{K\epsilon(t)} \quad (1 \leq j \leq m),$$

the map obtained by negating the sum in (5.8) sends $|u| \leq 1/2$ into $|u| \leq 1/4$ and has derivative bounded by $1/2$ there. It is a contraction. Its successive approximations converge uniformly on smaller parameter polydisks, so the unique fixed point is holomorphic in the s_j . The observable $(1+u)^q$ is bounded by a model-dependent constant M on the same domain.

Multivariate Cauchy estimates imply

$$|[s^k](1+u)^q| \leq MR(t)^{-|k|} = M(K\epsilon(t))^{|k|}.$$

By the coefficient calculation, the left side is $|t^{k \cdot \delta} B_k^{(q)}(L)|$. There are $\binom{n+m-1}{m-1}$ vectors with $|k| = n$. For $K\epsilon(t) \leq 1/2$, summing these bounds proves absolute convergence at $s_1 = \dots = s_m = 1$, and the tail after total degree N is bounded by a constant depending on N times $\epsilon(t)^{N+1}$. Multiplication by t^q gives (6.2).

At real $s_j = 1$, the fixed point is real and $1+u > 0$. It solves the original normalized equation and has $u \rightarrow 0$, so Theorem 3.1 identifies it with the required inverse. $\square$

This is a convergence theorem for the *exact finite model*, not for arbitrary formal transseries. It does not assert a universal numerical convergence threshold. In particular, the neighborhood can shrink when parameters change, and the estimates are not uniform as $\delta_* \rightarrow 0^+$.

6.2 The first excluded weight

Let $R > q$ be an output-power cutoff and set $B = R - q$. Define

$$\beta_+ = \min\{\beta \in \Gamma : \beta \geq B\}, \quad (6.3)$$

$$D_+ = \max \left\{ \sum_j k_j d_j : k \cdot \delta = \beta_+ \right\}. \quad (6.4)$$

Both are finite and computable. The set defining β_+ is nonempty because it contains an integer multiple of δ_* . More precisely,

$$\beta_+ \leq M_B := \left\lceil \frac{B}{\delta_*} \right\rceil \delta_* < B + \delta_* \quad (6.5)$$

for every $B > 0$. When B/δ_* is an integer, $M_B = B$. The implementation enumerates the closed weighted simplex up to M_B , so it sees the boundary even though the output cutoff is exclusive.

Theorem 6.2 (Next-weight remainder). *Let*

$$X_R(y) = t^q + \sum_{k \cdot \delta < R-q} t^{q+k \cdot \delta} B_k^{(q)}(\log t), \quad (6.6)$$

where the zero vector is omitted from the sum. Then

$$x(y) - X_R(y) = O\left(t^{q+\beta_+} \ell(t)^{D_+}\right). \quad (6.7)$$

In fact,

$$x(y) - X_R(y) = t^{q+\beta_+} C_{\beta_+}(\log t) + o\left(t^{q+\beta_+} \ell(t)^{D_+}\right). \quad (6.8)$$

If the aggregate C_{β_+} is nonzero, its actual degree and leading coefficient determine the leading asymptotic error.

Proof. Choose a fixed integer N with $(N+1)\delta_* > \beta_+$. The tail in (6.2) is

$$O\left(t^{q+(N+1)\delta_*} \ell(t)^{(N+1)d_*}\right) = o\left(t^{q+\beta_+} \ell(t)^{D_+}\right)$$

by the positive power gap. All indices with weight at most β_+ have degree at most N . Among the remaining finitely many indices of degree at most N , those at β_+ give exactly the polynomial C_{β_+} ; every larger weight is negligible by (2.5). This proves both statements. If $C_{\beta_+} \neq 0$, the same positive power-gap argument works with $\deg C_{\beta_+}$ in place of D_+ , so that polynomial also determines the leading asymptotic error. $\square$

The proof is more than the observation that a first omitted exponent exists. Logarithmic degrees grow with perturbation degree; an uncontrolled infinite tail cannot be bounded merely by looking at its smallest power. The finite-degree split and Proposition 6.1 supply the missing uniform control.

The package uses the safe degree D_+ before cancellations. It does not scan indefinitely to find the first *nonzero* omitted aggregate. Thus its remainder may be conservative in resonant or specially tuned examples, but it remains valid. When the first excluded power equals the cutoff and $D_+ > 0$, replacing its scale by a bare t^R is generally incorrect.

7 Worked examples and coefficient structure

7.1 The logarithmic example: a global real inverse

For $f(x) = x + x^2(1 + \log x)$,

$$f'(x) = 1 + x(3 + 2 \log x), \quad f''(x) = 5 + 2 \log x. \quad (7.1)$$

The derivative f' decreases up to $x = e^{-5/2}$ and increases thereafter. Its global minimum on $(0, \infty)$ is

$$m_0 = 1 - 2e^{-5/2} > 0. \quad (7.2)$$

Since $f(0^+) = 0$ and $f(x) \rightarrow \infty$, the function is a bijection of $(0, \infty)$ onto itself. Thus the local branch constructed above is also the unique global positive inverse.

Here $p = a = q = \delta_1 = 1$, $Q_1(L) = 1 + L$, and the expansion has the form

$$f^{-1}(y) = y + \sum_{n \geq 1} y^{n+1} A_n(\log y), \quad A_n(L) = \frac{(-1)^n}{n!} \prod_{j=2}^n (\mathcal{D} + n + j)(1 + L)^n. \quad (7.3)$$

For example,

$$A_2(L) = \frac{1}{2}(\mathcal{D} + 4)(1 + L)^2 = (1 + L) + 2(1 + L)^2 = 3 + 5L + 2L^2,$$

and

$$A_3(L) = -\frac{1}{6}(\mathcal{D} + 5)(\mathcal{D} + 6)(1 + L)^3.$$

These computations give (1.3). For any fixed $N \geq 0$,

$$f^{-1}(y) = y + \sum_{n=1}^N y^{n+1} A_n(\log y) + O\left(y^{N+2} \ell(y)^{N+1}\right). \quad (7.4)$$

The next weight is exactly $N + 1$, and its logarithmic degree is $N + 1$.

Catalan leading coefficients. The constant factors in (7.3) range from $n + 2$ to $2n$. Therefore

$$[L^n]A_n(L) = (-1)^n \frac{(2n)!}{n!(n+1)!} = (-1)^n \text{Cat}_n. \quad (7.5)$$

This identity is useful both conceptually and computationally. If one freezes the slowly varying factor at its leading logarithmic value, the dominant coefficients resemble inversion of a quadratic perturbation; (7.5) states the precise surviving highest-logarithm law. The lower logarithmic powers are not obtained by freezing the logarithm: they also contain the effect of its derivatives.

A triangular recurrence that checks the formula. Introduce a formal variable s independent of L , and put $U(s) = \sum_{n \geq 1} A_n(L)s^n$. The unit equation reads

$$U = -s(1 + U)^2(1 + L + \log(1 + U)). \quad (7.6)$$

It follows that

$$A_n(L) = -[s^{n-1}](1 + U_{n-1})^2(1 + L + \log(1 + U_{n-1})),$$

where $U_{n-1} = \sum_{j=1}^{n-1} A_j(L)s^j$. The right side depends only on earlier coefficients. The independent verification checks this recurrence against the differential-operator formula through $n = 8$.

Why this is not a Taylor series. The inverse extends continuously to zero and has right derivative 1. For $y > 0$, ordinary differentiation of the inverse gives

$$(f^{-1})''(y) = -\frac{5 + 2\log(f^{-1}(y))}{(1 + f^{-1}(y)(3 + 2\log(f^{-1}(y))))^3} \rightarrow +\infty. \quad (7.7)$$

Hence it does not have a finite second right derivative at zero, much less an analytic Taylor expansion there. The singularity is compatible with, and directly visible in, the $y^2 \log y$ term.

7.2 A single arbitrary real power

Consider

$$F_{\alpha,c}(x) = x + cx^\alpha, \quad \alpha > 1, \quad c \in \mathbb{R}. \quad (7.8)$$

Its positive inverse exists near zero for every fixed real c . The general formula reduces to

$$F_{\alpha,c}^{-1}(y) = y + \sum_{n \geq 1} (-c)^n b_n(\alpha) y^{1+n(\alpha-1)}, \quad (7.9)$$

$$b_n(\alpha) = \frac{(n\alpha)^{n-1}}{n!} = \frac{1}{n} \binom{n\alpha}{n-1} = \frac{1}{1+n(\alpha-1)} \binom{n\alpha}{n}. \quad (7.10)$$

The falling factorial is a finite product, so the coefficient formula needs no special-function evaluation and is meaningful for every real $\alpha > 1$.

The first coefficients are

$$\begin{aligned} F_{\alpha,c}^{-1}(y) = y - cy^\alpha + c^2\alpha y^{2\alpha-1} - \frac{c^3\alpha(3\alpha-1)}{2} y^{3\alpha-2} \\ + \frac{c^4\alpha(4\alpha-1)(2\alpha-1)}{3} y^{4\alpha-3} + \mathcal{O}(y^{5\alpha-4}). \end{aligned}$$

At $\alpha = \sqrt{2}$ and $c = 1$, the cubic correction coefficient is $-(6 - \sqrt{2})/2$, reproducing the question. The exponent increment $\sqrt{2} - 1$ is irrational, but positive; support enumeration is simply enumeration of its nonnegative integer multiples.

For $c = 1$, $F'_{\alpha,1}(x) = 1 + \alpha x^{\alpha-1} > 0$ on the entire positive axis. Thus the inverse in this case is globally unique. For negative c , the construction remains local and makes no corresponding global claim.

An exact convergence radius in the small parameter. Put $s = cy^{\alpha-1}$ and $F_{\alpha,c}^{-1}(y) = yW(s)$. Then $W + sW^\alpha = 1$, $W(0) = 1$, and

$$W(s) = 1 + \sum_{n \geq 1} (-1)^n b_n(\alpha) s^n.$$

Its Taylor radius is

$$R_\alpha = \frac{(\alpha - 1)^{\alpha-1}}{\alpha^\alpha}. \quad (7.11)$$

To verify this, write

$$b_n(\alpha) = \frac{\Gamma(n\alpha + 1)}{\Gamma(n + 1)\Gamma(n(\alpha - 1) + 2)}.$$

Stirling's formula gives

$$b_n(\alpha) \sim \frac{\sqrt{\alpha}}{\sqrt{2\pi}(\alpha - 1)^{3/2}} R_\alpha^{-n} n^{-3/2}. \quad (7.12)$$

The root test proves (7.11). The same value is suggested by the simultaneous equations $W + sW^\alpha = 1$ and $1 + \alpha sW^{\alpha-1} = 0$, which give $W = \alpha/(\alpha - 1)$ and $s = -R_\alpha$. Thus (7.9) is not merely formal: it converges whenever $|c|y^{\alpha-1} < R_\alpha$. At $\alpha = 2$ this specializes to the familiar Catalan radius $1/4$.

7.3 Mixed logarithmic and irrational generators

Let

$$F(x) = x + bx^{\sqrt{2}} + cx^{3/2}(1 + \log x), \quad b, c \in \mathbb{R}.$$

The two relative weights are $\delta_1 = \sqrt{2} - 1$ and $\delta_2 = 1/2$. Below output power 2, the inverse is

$$\begin{aligned} F^{-1}(y) = y - by^{\sqrt{2}} - cy^{3/2}(1 + L) + \sqrt{2}b^2y^{2\sqrt{2}-1} \\ + bc y^{\sqrt{2}+1/2} \left[\left(\sqrt{2} + \frac{3}{2} \right) (1 + L) + 1 \right] + O(y^2 \ell(y)^2). \end{aligned} \quad (7.13)$$

The mixed coefficient corresponds to $\mathbf{k} = (1, 1)$ and is obtained by one application of $\mathcal{D} + \sqrt{2} + 3/2$ to $bc(1 + L)$. This example shows why neither ordinary integer order nor total perturbation degree alone is the requested output ordering. A single weighted support calculation handles both types of term.

7.4 Resonances

For

$$F(x) = x + Ax^{3/2} + Bx^2,$$

the weight 1 has two representations: $2(1/2)$ and 1 . The inverse begins

$$\begin{aligned} F^{-1}(y) = y - Ay^{3/2} + \left(\frac{3}{2}A^2 - B \right) y^2 \\ + \left(-\frac{21}{8}A^3 + \frac{7}{2}AB \right) y^{5/2} + O(y^3). \end{aligned} \quad (7.14)$$

The coefficient of y^2 is not a choice between two candidate answers: it is their sum. It may vanish, for example when $B = 3A^2/2$. That cancellation does not alter the semigroup, but it changes the nonzero support and can make the generic error envelope conservative.

7.5 A non-unit leading power

For $F(x) = x^2 + x^3$, use $v = x^2$, $q = 1/2$, and $t = v + v^{3/2}$. Applying the observable formula, rather than returning v itself, gives

$$F^{-1}(y) = y^{1/2} - \frac{1}{2}y + \frac{5}{8}y^{3/2} - y^2 + \frac{231}{128}y^{5/2} + O(y^3). \quad (7.15)$$

The derivative of the forward map tends to zero, so a residual and an inverse error no longer have the same power. A candidate with inverse error of order y^3 has forward residual of order $y^{7/2}$. This is the reason for the cutoff conversion in the residual checker.

For a positive leading coefficient $a \neq 1$, the natural coordinate is $t = y/a$. The package deliberately leaves $(y/a)^\gamma$ and $\log(y/a)$ in this form. It does not use the complex-unsafe blanket transformation of powers often associated with `PowerExpand`. Under $a > 0$ and $y > 0$, conversion to powers of y and polynomials in $\log y - \log a$ is mathematically legitimate, but not needed internally.

8 Residual certificates and uncertain input expansions

8.1 The correct transport of a residual

Let $x(y)$ denote the exact inverse and let $X(y) > 0$ be a candidate with $X(y)/t^q \rightarrow 1$. Define the exact normalized residual

$$E(y) = \frac{f(X(y))}{a} - t. \quad (8.1)$$

Proposition 8.1 (Residual-to-inverse transport). *For a fixed model (2.1),*

$$|X(y) - x(y)| \leq Ct^{q-1}|E(y)| \quad (8.2)$$

for sufficiently small positive y . More precisely, at points where the residual is nonzero,

$$X(y) - x(y) = \frac{1}{p}t^{q-1}E(y)(1 + o(1)). \quad (8.3)$$

Thus a residual of order $O(t^A \ell(t)^d)$ gives inverse error $O(t^{A+q-1} \ell(t)^d)$.

Proof. The mean value theorem gives $f(X) - f(x) = f'(\xi)(X - x)$ for a point ξ between X and x . Both endpoints are asymptotic to t^q , hence so is ξ . Equation (3.2) gives

$$f'(\xi) = ap t^{q(p-1)}(1 + o(1)) = ap t^{1-q}(1 + o(1)).$$

Divide the residual identity by this derivative. □

An inverse jet retaining powers below R should therefore have vanishing forward residual below

$$C = R + 1 - q. \quad (8.4)$$

For $p = 1$ this equals R , but for a square-root inverse it equals $R + 1/2$. Testing a residual at an unconverted cutoff can be too weak or unnecessarily strong.

8.2 Independent finite composition

Write the candidate as $X = t^q(1 + U)$, where U is a finite power-log jet with strictly positive least power ν . An original block transforms as

$$X^\alpha P(\log X) = t^{q\alpha}(1 + U)^\alpha P(qL + \log(1 + U)). \quad (8.5)$$

Introduce an ordinary auxiliary variable z and expand the analytic unit

$$H(z, L) = (1 + z)^\alpha P(qL + \log(1 + z))$$

at $z = 0$. The term containing U^j has power at least $j\nu$. Therefore only finitely many j can contribute below a residual cutoff C : it suffices to compute through $\lfloor (C - q\alpha)/\nu \rfloor$, then discard the boundary if necessary. Sparse multiplication of U is truncated at each step. Because all powers of U are positive, an already discarded term can never return to a lower power after another multiplication.

This composition procedure does not use the inverse coefficient formula. It is the algorithm implemented by `InverseResidual`. It returns a finite *residual jet*, not the entire nonlinear expression $f(X) - y$. In particular, a returned residual expression of zero means “all retained coefficients vanish,” not “the finite approximation is an exact inverse.” The function checks the stored “Blocks” field; editing only a separate “Expression” field does not change what is checked.

8.3 Numerical a posteriori bounds for the first example

For (1.1), the global lower bound (7.2) is available. Every positive trial point X obeys

$$|X - f^{-1}(y)| \leq \frac{|X + X^2(1 + \log X) - y|}{1 - 2e^{-5/2}}. \quad (8.6)$$

If the exact residual is positive, the inverse lies between $X - |f(X) - y|/m_0$ and X ; if it is negative, it lies between X and $X + |f(X) - y|/m_0$. One may intersect these intervals with $(0, \infty)$.

Such a statement is a numerical certificate only when the residual is itself rigorously enclosed, for example with appropriately controlled interval arithmetic. A high-precision floating-point computation alone is a diagnostic, not a certified enclosure. The delivered package does not claim to manufacture interval certificates or explicit constants for its asymptotic big- O metadata.

More generally, suppose a derivative lower bound $m > 0$ is verified on an interval containing $X - \rho$ and $X + \rho$, and $|f(X) - y| \leq m\rho$. Integration of the derivative bound gives opposite weak signs at the two endpoints. Continuity provides a root in the interval, and strict monotonicity makes it unique. This separates a genuine root-existence certificate from a merely small residual.

8.4 Transporting an input remainder

Suppose the actual function is $F = f + r$, where f is the finite model,

$$r(x) = O(x^\lambda \ell(x)^M), \quad \lambda > p, \quad (8.7)$$

and $F'(x) = apx^{p-1}(1 + o(1))$. The latter condition follows, for example, from $r'(x) = o(x^{p-1})$. Let x_F and x_f be the positive inverse branches. Evaluating F at x_f and applying the same mean-value argument gives

$$x_F(y) - x_f(y) = O(t^{q\lambda+q-1}\ell(t)^M). \quad (8.8)$$

Thus input uncertainty and algorithmic truncation must both be included. If the missing forward term is of order $x^{p+\eta}$, its possible effect on the inverse begins at power $q + \eta/p$.

The derivative condition is not cosmetic. The function

$$F(x) = x + x^2 \sin(x^{-3})$$

satisfies $F(x)/x \rightarrow 1$, but its derivative changes sign arbitrarily close to zero. Mere relative closeness to a monotone leading term does not supply a single monotone inverse germ. The finite power-log model avoids this problem because its derivatives have controlled scales.

Example: an elementary function first reduced to a jet. Near zero,

$$x + \sin x = 2x - \frac{x^3}{6} + \frac{x^5}{120} + O(x^7),$$

with the corresponding derivative control. Inverting the displayed model and transporting the input remainder gives

$$(x + \sin x)^{-1}(y) = \frac{y}{2} + \frac{y^3}{96} + \frac{y^5}{1920} + O(y^7).$$

The package rejects an unexpanded `Sin[x]` in its exact-model front end, but accepts the displayed polynomial model. The theorem explains precisely what one may conclude about the original elementary function; applying `Normal` to a forward series without recording its error would not explain this by itself.

9 Algorithms and implementation contracts

9.1 Inversion algorithm

The principal routine implements the following finite calculation.

1. Parse and combine the exact input blocks. Check real coefficients, positive leading coefficient, positive leading power, and exact ordered numerical exponents. Normalize by (2.2).
2. Set $B = R - q$ and compute the frontier bound M_B in (6.5). Enumerate the integer vectors $\mathbf{k} \cdot \delta \leq M_B$, pruning recursively by the remaining weight budget. Impose explicit resource limits.
3. For vectors with $0 < \mathbf{k} \cdot \delta < B$, form $Q^{\mathbf{k}}$ and apply the $n - 1$ first-order operators in (5.2). Combine equal exponents and sort them by exact comparison.
4. Find β_+ and D_+ among the excluded frontier vectors. Return the expression, coefficient blocks, model data, branch convention, and remainder scale from Theorem 6.2.

All indices below the requested cutoff are included because the frontier bound is at least B . No index beyond that bound can have a smaller excluded weight. The coefficient theorem proves the formula applied to each included vector; exact grouping proves resonance handling; the remainder theorem proves the meaning of the returned error scale. These observations establish the mathematical correctness of the algorithm independently of any particular CAS evaluation.

9.2 Representation and differentiation

Internally, a block is a pair

```
{exponent, polynomialInLogCoordinate}
```

where `LogCoordinate` is a protected public indeterminate. A coefficient operator is implemented by repeated updates of the form

```
poly = Expand[D[poly, LogCoordinate] +
              (beta + q + j) poly];
```

The numerical variable is not present in these polynomial calculations. Only when constructing an output expression is `LogCoordinate` replaced by $\text{Log}[y/a]$ and the block multiplied by $(y/a)^{\text{exponent}}$.

The parser replaces only the literal logarithm $\text{Log}[x]$ by a fresh indeterminate, expands finite algebraic sums, and extracts factors that are powers of the input variable. It deliberately does not rewrite arbitrary nested logarithms or fractional powers of sums. Positivity of x makes many additional identities possible, but a controlled parser should not obtain them through indiscriminate complex power rules.

9.3 Exact order and failure behavior

The comparison routine tries to prove $\alpha - \beta = 0, < 0$, or > 0 using exact simplification under the supplied assumptions. If none can be established, it returns a `Failure`. There is no decimal fallback for support ordering. Likewise, integer bounds for enumeration must be established as exact integers.

The implemented front end requires exponent constants such as integers, rationals, radicals, or other exact real numerical constants that the underlying CAS can compare. It does not accept unrestricted symbolic exponents even when a partial inequality, such as $\alpha > 1$, is known. That restriction is an implementation boundary, not a restriction of the mathematical theorem. Coefficients may be symbolic if their reality and the leading coefficient's positivity follow from `Assumptions`.

Unsupported input, unproved sign conditions, an invalid cutoff, and exceeded enumeration limits return a structured `Failure`; they do not produce a guessed branch or a silently truncated coefficient set. The package reserves `LogCoordinate` and requires distinct, unassigned input and output symbols.

9.4 Cost and current engineering limits

Let K be the number of enumerated vectors and $N_{\max} = \lfloor M_B/\delta_* \rfloor$. A crude bound is

$$K \leq \prod_{j=1}^m \left(1 + \left\lfloor \frac{M_B}{\delta_j} \right\rfloor \right). \quad (9.1)$$

For fixed weights and fixed m , this is polynomial growth in the cutoff, though the dimension m matters substantially. A coefficient of total degree n requires $n - 1$ first-order polynomial operators and has logarithmic degree at most nd_* .

The reference implementation prioritizes transparency and exact comparisons. It uses simple list-based aggregation, so exponent merging can entail quadratically many comparisons in the number of blocks. It is not an optimized high-order transseries engine. Large logarithmic degrees can also cause substantial exact-expression growth. Defaults limit total frontier order to 64 and enumerated vectors to 100000; users can change these bounds explicitly. The limits are safety controls, not performance promises.

Useful future optimizations include canonical algebraic-number keys, cached polynomial powers, memoized operator products, direct aggregation by a rational lattice when one exists, and Newton lifting in the same sparse jet algebra. None is needed to make the present finite calculation well-defined or to obtain the requested examples.

10 Wolfram Language interface

10.1 RealInverseExpansion

The interface is

```
RealInverseExpansion[expr, {x, y}, cutoff,
  Assumptions -> True,
  "MaxMultiIndices" -> 100000,
  "MaxTotalOrder" -> 64]
```

It accepts a finite sum of terms $x^\alpha P(\log x)$ with a positive pure-power leading term. Inputs containing approximate real numbers are rejected; construct the expansion exactly, then substitute numerical parameters. The cutoff must be an exact real numerical constant strictly greater than $q = 1/p$.

The principal result fields are as follows.

Field	Meaning
"Expression"	The finite inverse approximation, evaluated in the coordinate $t = y/a$.
"Blocks"	Ordered pairs $\{q + \beta, C_\beta(L)\}$, using LogCoordinate for L . These are the data checked by the residual routine.
"RemainderScale"	A scale $t^{q+\beta_+} \ell(t)^{D_+}$ such that the error is big- O of that scale for the exact model. It is zero for a pure monomial.
"RemainderExponent"	$q + \beta_+$, or Infinity for the exact pure-monomial case.
"RemainderLogDegree"	The safe degree D_+ .
"Cutoff"	The exclusive output-power cutoff.
"LeadingInversePower"	$q = 1/p$.
"ScaledVariable"	The expression y/a .
"Model"	Original blocks, leading power and coefficient, normalized corrections, and the variable names.
"Assumptions"	The coefficient and parameter assumptions used for exact simplification. Positive y is part of the branch contract.
"Branch"	The positive-real convention $x/(y/a)^{1/p} \rightarrow 1$ as $y \rightarrow 0^+$.
"Exact"	True for the pure-monomial special case; False otherwise. It does not indicate numerical precision.
"EnumeratedMultiIndices"	Number of nonzero frontier vectors examined.
"ContributingMultiIndices"	Number of included vectors before resonant grouping and cancellation.

The expression is not wrapped in ConditionalExpression; its domain and branch are part of the documented result contract. It should be evaluated only for $y > 0$ and parameter values satisfying the assumptions. All asymptotic assertions concern $y \rightarrow 0^+$.

For example, the first nontrivial logarithmic output is represented by

```
{{1, 1}, {2, -1 - LogCoordinate}}
```

with remainder exponent 3 and logarithmic degree 2. For the second example, $4 \sqrt{2} - 3$ excludes the fifth block exactly; no floating-point approximation to that cutoff is required.

10.2 InverseResidual

Use

```
InverseResidual[result]
InverseResidual[result, explicitResidualCutoff]
InverseResidual[result, Automatic,
  "MaxCompositionOrder" -> 128]
```

The default cutoff is (8.4). The result stores the coefficient pairs in "ResidualBlocks". Its expression fields are "NormalizedResidualExpression" and "ResidualExpression"; the latter is the normalized residual jet multiplied by a . It also returns "Cutoff" and the boolean "VanishingBelowCutoff".

The routine expands only ordinary analytic units in an auxiliary variable, using `Series` where ordinary Taylor series are appropriate. It does not ask `Series` to solve an inverse at the singular endpoint. This distinction also explains why using an association of exact blocks is preferable to forcing the answer into `SeriesData`. The documented `SeriesData` representation has a single evenly stepped power index [7]; the package needs an arbitrary exactly ordered positive semigroup and coefficient polynomials in a logarithm.

A failed composition identity is informative: a wrong coefficient, an omitted resonant contribution, or an incorrect logarithmic shift leaves a nonzero residual block. The test suite deliberately corrupts a stored coefficient to ensure that the checker is designed to detect such an error rather than merely returning an expected status.

10.3 LagrangeInverseTruncation

The low-level helper is

```
LagrangeInverseTruncation[h, {x, t}, n]
LagrangeInverseTruncation[h, {x, t}, n,
  "OutputPower" -> q]
```

Here h is the correction in $x + h(x)$, not the entire forward function. The helper returns

$$t^q + \sum_{k=1}^n \frac{q(-1)^k}{k!} \left(\frac{d}{dx} \right)^{k-1} (x^{q-1} h(x)^k) \Big|_{x=t}. \quad (10.1)$$

For $q = 1$ it is the near-identity inverse truncation. It accepts broader symbolic expressions and symbolic exponent parameters because it does not enumerate or order their output support. It does *not* prove $h = o(x)$, derivative estimates, convergence at the auxiliary parameter 1, or an asymptotic remainder. Those hypotheses remain the caller's responsibility. This separation prevents a general formal identity from being misrepresented as a universal certified asymptotic solver.

10.4 Numerical evaluation and use with parameters

A typical exact-to-numeric workflow is

```
Clear[x, y, a, c];
r = RealInverseExpansion[a x + c x^2, {x, y}, 5,
  Assumptions -> a > 0 && Element[c, Reals]];
N[r["Expression"] /. {a -> 3, c -> -2, y -> 10^-8}, 80]
```

Do not start with $10.^{-8}$ and later raise its displayed precision: that does not recover digits that were never present. Likewise, an approximate exponent can destroy the exact identity of resonant powers. For numerical comparisons, sufficient guard digits should be used when subtracting a small residual or an error much smaller than the inverse itself.

The directory can be loaded directly with `Get`; no installation, network service, or third-party Wolfram add-on is required for execution in a suitable kernel. An optional `PacletInfo.wl` and conventional `Kernel/init.m` are supplied. The declared compatibility target is Wolfram Language 12.0 or later; this is an implementation target, not a claim of executed cross-version testing.

11 Verification and reproducibility

11.1 What was actually executed

The delivered independent verification script ran under Python 3.13.5, SymPy 1.14.0, and mpmath 1.3.0, with 180 decimal digits for numerical comparisons. Its recorded result is **66 checks passed out of 66**. The result files identify each check individually.

The checks comprise eight logarithmic coefficients tested in three ways (implicit recursion, direct differentiation, and the Catalan leading-log identity); eight irrational-power coefficients checked by a separate implicit recursion; eight independent composition checks for different models; four support, remainder, resonance, and corruption checks; four static delimiter checks; and eighteen high-precision numerical comparisons. The composition models include mixed irrational/logarithmic support, resonance, nonlinear leading powers, nonunit leading coefficients, a pure monomial, and quadratic logarithmic coefficients.

The unit-series recurrences used independently are worth recording. If $A(s) = 1 + \sum_{n \geq 1} a_n s^n$ and $A(s)^\theta = \sum_{n \geq 0} b_n s^n$, then $AB' = \theta BA'$ gives

$$b_0 = 1, \quad nb_n = \sum_{k=1}^n ((\theta + 1)k - n) a_k b_{n-k}. \quad (11.1)$$

If $\log A(s) = \sum_{n \geq 1} \ell_n s^n$, the identity $A(\log A)' = A'$ gives

$$\ell_n = a_n - \frac{1}{n} \sum_{k=1}^{n-1} k \ell_k a_{n-k}. \quad (11.2)$$

These recurrences do not call the inverse coefficient formula. They provide an algebraically independent route to the unit compositions and implicit coefficient solutions used in the checks.

11.2 Numerical results

Table 2 reports relative errors against independently computed positive roots. Here N is the number of correction orders, so the leading term is additional. For the logarithmic example, retaining N corrections means powers through y^{N+1} ; for the irrational example, it means powers through $y^{1+N(\sqrt{2}-1)}$.

The slower improvement for the irrational example at a fixed y is expected: its small parameter is $y^{\sqrt{2}-1}$ rather than approximately $y |\log y|$. The table is a numerical consistency check, not a substitute for the error proofs and not a rigorous floating-point enclosure. Full values, absolute errors, and residual magnitudes are retained in `Verification/results.json`.

Example	y	$N = 2$	$N = 4$	$N = 6$
Logarithmic	10^{-2}	1.746×10^{-4}	1.242×10^{-6}	1.039×10^{-8}
Logarithmic	10^{-4}	2.408×10^{-9}	1.150×10^{-14}	6.630×10^{-20}
Logarithmic	10^{-8}	2.478×10^{-20}	5.850×10^{-33}	1.674×10^{-45}
Irrational	10^{-2}	6.806×10^{-3}	4.740×10^{-4}	3.852×10^{-5}
Irrational	10^{-4}	2.414×10^{-5}	3.764×10^{-8}	6.802×10^{-11}
Irrational	10^{-8}	2.625×10^{-10}	1.994×10^{-16}	1.753×10^{-22}

Table 2: Observed relative errors, computed at 180 decimal digits and rounded for display. The reference roots were computed independently by Newton iteration. These are numerical diagnostics, not interval certificates.

11.3 What was not executed

The archive supplies **34 native Wolfram verification tests** and a `wolframscript` runner. Those tests were **not executed** in the preparation environment. The available Wolfram service returned HTTP 404, and no local Wolfram kernel was installed. Static delimiter checks are not a parser, a type checker, or kernel execution. The Python reference implementation verifies the mathematics and its own calculations; it does not establish runtime correctness of the Wolfram source.

This distinction is part of the delivery record, not an omitted qualification. The package has been written and reviewed against the specified Wolfram semantics, and the native tests are intended to make an actual kernel run straightforward and auditable. Until such a run is recorded, compatibility and runtime behavior should not be described as empirically verified. No current result of `AsymptoticSolve`, `InverseSeries`, or the historical `InverseFunction` calls is claimed to have been reproduced.

11.4 Commands to reproduce the calculations and document

From the extracted top-level directory:

```
python Verification/verify.py
wolframscript -file Tests/run.wls
cd Article
pdflatex -interaction=nonstopmode -halt-on-error real_inverse_asymptotics.tex
pdflatex -interaction=nonstopmode -halt-on-error real_inverse_asymptotics.tex
pdflatex -interaction=nonstopmode -halt-on-error real_inverse_asymptotics.tex
```

The Python script needs SymPy and mpmath. Its output records software versions and rewrites the independent results files. The Wolfram runner prints the kernel version and test counts and writes a native test report; no such report is pre-populated in this archive. The article builds with a standard TeX Live installation containing the packages named in its preamble. Its source and the numerical-table fragment are included.

12 Extensions and precise boundaries

12.1 What the theory and implementation already cover

Within the exact finite-model class, all coefficients are covered, not just the displayed examples. Arbitrary positive real leading powers, multiple exact irrational corrections, logarithmic polynomials of any finite degree, symbolic real coefficients under assumptions, and resonant exponent sums are handled by the same formula. Any desired finite exclusive power cutoff determines a finite mathematical computation.

The implementation can stop early only with an explicit resource or undecidability failure, rather than silently claiming the requested order.

The output class is a finite-generator power–logarithmic subclass of broader transseries calculi; Edgar’s exposition gives context for those larger structures [8]. No claim of a complete implementation of the full logarithmic-exponential transseries field is made.

12.2 Leading logarithms and other dominant cores

A function whose leading term contains a logarithm requires another normalization. For instance,

$$y = -x \log x, \quad x \rightarrow 0^+,$$

has the small positive inverse

$$x = \exp(W_{-1}(-y)) = -\frac{y}{W_{-1}(-y)}.$$

To derive this, put $u = \log x$ and solve $ue^u = -y$; the branch $u \rightarrow -\infty$ is W_{-1} . This inverse brings in inverse logarithms and iterated logarithms. It is not a finite power–log perturbation of a pure positive monomial in the sense of (2.1). The present front end rejects such input instead of choosing the wrong Lambert branch. A future dominant-core layer could supply the appropriate exact inverse and transport corrections around it, in the spirit of the repository’s core-inversion architecture [4].

12.3 Symbolic exponents and parameter chambers

If an exponent is a parameter, support order can change with that parameter. For example, comparing $2\delta_1$ with δ_2 requires knowing on which side of $2\delta_1 = \delta_2$ the parameters lie. A fully symbolic ordered solver should decompose parameter space into chambers and treat the resonance boundaries separately. Merely knowing that both weights are positive is insufficient to choose one ordered list of the first few blocks. The derivative helper remains useful in this situation, but its perturbation grading is not an automatic sorted asymptotic grading.

12.4 Infinite supports, inverse logarithms, and flat terms

Finitely many strictly positive weights are what make every cutoff a finite task. If infinitely many generators approach zero, a bounded power interval can contain infinitely many independent contributions. Different support hypotheses and truncation semantics are then needed. Similarly, allowing rational or asymptotic functions of L rather than polynomials can create infinitely many logarithmic orders at a fixed power. Those extensions require a second, explicit coefficient-scale truncation, not just removal of the polynomial check in the parser.

A function and its perturbation by a term such as $e^{-1/x}$ may share all power–log coefficients while having different exact inverses. Under the derivative conditions of Section 8.4, their inverse difference is correspondingly flat. Thus a formal expansion alone does not uniquely determine an arbitrary smooth inverse. The convergence theorem here identifies the inverse because the finite forward model is specified exactly, rather than only up to a flat error.

12.5 A possible formal verification boundary

The finite coefficient engine has a relatively small mathematical trust boundary: polynomial arithmetic over exact coefficients; the block operator identity; finite weighted-support enumeration; exact exponent comparisons; and the finite composition identity. These pieces are natural candidates for independent formal verification. The analytic branch, convergence, and remainder theorems form a separate layer that

connects those finite computations to real functions. Formalizing only the coefficient identities would not by itself certify that analytic connection, a distinction emphasized in the repository's own status records [5].

13 Conclusion

The right replacement for a Taylor expansion of an unspecified complex inverse is an expansion of a *specified positive real germ* on its correct ordered scale. For the two questions at hand, the complete solution is supplied by local Lagrange inversion around a positive point, followed by a uniform small-parameter argument. Polynomial logarithmic coefficients are managed by constant-coefficient differential operators; irrational powers are managed by a locally finite positive semigroup. Neither feature requires numerical guessing or a symbolic global inverse.

The resulting package keeps the three logically different tasks separate: computing exact coefficient blocks, certifying their finite composition identity, and proving the asymptotic meaning of the omitted tail. That separation is what makes the original real-branch examples, their mixed generalizations, and nonunit leading powers accessible to the same explicit algorithm.

A Additional logarithmic coefficients

The polynomials A_n in (7.3) through $n = 4$ appear in (1.3). The next two are

$$\begin{aligned} A_5(L) &= -42L^5 - \frac{3727}{12}L^4 - \frac{5365}{6}L^3 - 1256L^2 - \frac{5177}{6}L - \frac{2789}{12}, \\ A_6(L) &= 132L^6 + \frac{5981}{5}L^5 + \frac{52937}{12}L^4 + \frac{16979}{2}L^3 \\ &\quad + 9007L^2 + \frac{30019}{6}L + \frac{22769}{20}. \end{aligned}$$

Keeping these through $n = 6$ gives error $O(y^8 \ell(y)^7)$. Coefficients through $n = 8$ are recorded in `Verification/results.json`; the operator formula produces any further fixed order. The leading coefficients -42 and 132 are $(-1)^5 \text{Cat}_5$ and $(-1)^6 \text{Cat}_6$, respectively.

For numerical evaluation it is often more efficient to evaluate each coefficient polynomial by Horner's rule than to expand the entire expression in the numerical variable. The supplied symbolic package prioritizes exact readable output, while arbitrary-precision evaluation can be applied after the exact substitution.

B Archive organization and trust record

```
RealInverseAsymptotics/  
  README.md  
  PacletInfo.wl  
  Kernel/  
    RealInverseAsymptotics.wl  
    init.m  
  Examples/  
    examples.wl  
  Tests/  
    RealInverseAsymptotics.wlt  
    run.wls  
  Verification/  
    verify.py  
    results.json  
    results.txt  
  Article/  
    real_inverse_asymptotics.tex  
    real_inverse_asymptotics.pdf  
    numerical_table.tex  
    build.sh  
  MANIFEST.sha256
```

The manifest records checksums for delivered files other than the manifest itself. No external paper, repository volume, font file, or proprietary kernel is redistributed. The article is self-contained apart from ordinary LaTeX package dependencies; its proofs do not depend on having access to the linked repository.

The Wolfram tests cover the two original examples, logarithmic error metadata, exclusive irrational cutoffs, Catalan and resonant cases, positive power cores, symbolic coefficients and slopes, residual composition, deliberate corruption, the general derivative helper, invalid inputs, resource limits, and one high-precision root comparison. Their expected outputs are specifications awaiting an actual native kernel run, not fabricated execution receipts.

References

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- [2] NIST Digital Library of Mathematical Functions, §1.10(vii), *Inverse Functions*, Lagrange Inversion Theorem and Extended Inversion Theorem. <https://dlmf.nist.gov/1.10#vii>. Consulted 7 September 2026.
- [3] Ira M. Gessel, *Lagrange Inversion*, Journal of Combinatorial Theory, Series A **144** (2016), 212–249. <https://arxiv.org/abs/1609.05988>. The arXiv version’s Theorem 2.1.1 states the coefficient form used here.
- [4] Vladimir Reshetnikov et al., ProveIt repository, *Series and transseries*, group README. <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/series-and-transseries>. Consulted 7 September 2026. Context used: comparison of operator and coefficient inversion, dominant-core architecture, and residual transport.
- [5] Vladimir Reshetnikov et al., ProveIt repository, *Transseries: the polynomial–logarithmic calculus and its inversions*, canonical-volume README. https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/series-and-transseries/Transseries_And_Inversion. Consulted 7 September 2026. Context used: scoped formalization inventory and explicit boundaries between formal identities and analytic claims.
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- [8] G. A. Edgar, *Transseries for Beginners*, Real Analysis Exchange **35** (2009/2010), 253–309. <https://arxiv.org/abs/0801.4877>. The real grid-based viewpoint provides broader context; the finite-model proofs in this article are given independently.